Ethical Reasoning in Large Language Models: A Comparative Study of Gemini Flash and Gemini Flash-Lite

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Abstract

This paper asks whether two models from the same family, Google's Gemini 2.5 Flash and Gemini 2.5 Flash-Lite, systematically differ in responses to morally ambiguous prompts. We collected 60 responses to 30 ethical scenarios across six categories and analysed each with two complementary classifiers: a keyword-based classifier and a pre-trained transformer sentiment classifier. The main finding is that the two models are cautious in different ways: Flash-Lite hedges more in language, while Flash uses a more emotionally concerned tone. The gap is largest in medical ethics, resource allocation, and privacy vs. safety, and smallest on abstract trolley problems. The contribution is also methodological: a reproducible pipeline that other studies can reuse.

Keywords: Large language models, ethical reasoning, Gemini, sentiment analysis, keyword classification

1 Introduction

Large language models are now used in applications where their responses to ethical questions matter: medical decision support, content moderation, education, policy work. Considerable work compares different providers on such questions [1, 2]. Far fewer studies ask a related but different question: how do two models from the same provider and same family differ from each other when they are presented as variants with different speed, cost, and capacity profiles?

This question matters in practice. Teams routinely choose between a smaller lite model and a larger one in the same family. Scientifically, comparing two models within the same family is also cleaner than any cross-provider comparison, because provider and family are held constant even though exact training and alignment details are not fully public.

We study this question for the Gemini 2.5 family using two classifiers, both from natural language processing. The first is a keyword-based classifier we wrote from scratch, using text normalisation, tokenisation, regex-based sentence segmentation, and lexicon matching to count hedge words, refusal phrases, direct commitments, and two-sided framings, and to assign each response one strategy label (refusal, hedged, balanced, direct, or mixed). The second is a pre-trained transformer sentiment classifier from the BERT family [3, 4] that returns positive, negative, and neutral probabilities and a discrete label. In line with the course brief, the goal is not to maximise any metric but to study a focused research question carefully. The main empirical finding is that the two models are cautious in different ways: Flash-Lite hedges more in language, while Flash uses a more emotionally concerned tone.

2 Research Question and Methodology

2.1 Research Question

The question we ask is a limited one: do two models from the same family systematically respond differently to the same morally ambiguous prompt, and what linguistic, strategic, and emotional patterns characterise those differences?

By family we mean models from the same provider, broadly shared training data and alignment, differing mainly in size. Morally ambiguous prompts are ones where reasonable people would disagree too. The question is limited in its ambition. We do not try to determine which model is more ethical, and we do not judge whether any answer is right. We just ask if the answers are different, and how.

2.2 Dataset of Prompts

We wrote thirty prompts, divided into six categories of five prompts. The categories cover different types of ethical tension: trolley dilemmas (utilitarian vs. deontological pressure), medical ethics (end-of-life, autonomy, paternalism, confidentiality), privacy vs. safety (surveillance, informing on others), honesty vs. kindness (small lies that protect feelings), free speech vs. harm (platforming, moderation, contested speech), and resource allocation (zero-sum trade-offs). Each prompt is a short (one to three sentence), standalone real dilemma and asks the model to take a position. The full dataset can be found in the project repository at `data/prompts.json`.

2.3 Models Under Comparison

We compare two models from Google's Gemini 2.5 family: the standard cost-efficient **Gemini 2.5 Flash**, and the smaller, faster, cheaper **Gemini 2.5 Flash-Lite** meant for high-throughput use. Both models belong to the Gemini 2.5 family and are presented as variants with different speed, cost, and capacity profiles. This makes the comparison cleaner than a cross-provider comparison, although the exact training and alignment details are not fully public. Both models are available through Google's free-tier API, so the study is reproducible at no cost. We query both models with temperature 0.7 and a maximum output length of 2048 tokens. Temperature 0.7 gives moderate creativity: deterministic responses (temperature 0) would be too fixed for ethical reasoning, and higher temperatures introduce too much noise for consistent analysis. The maximum output length is generous enough for responses to develop fully.

2.4 Analysis Pipeline

The pipeline is designed to be simple and transparent. Two classifiers independently classify each response. The first is a keyword-based classifier we wrote from scratch, and the second is a pre-trained transformer sentiment classifier. They each provide both a quantitative output and a discrete categorical label, which are two complementary views of the same response.

For each response the keyword classifier computes length features (word count, sentence count, character count) and four keyword counts: hedge count, the number of hedge expressions like "however", "may", "might"; refusal count, the number of refusal phrases like "I cannot", "as an AI"; direct count, the number of commitment phrases like "the answer is", "I believe"; and balanced count, the number of two-sided phrases like "on one hand", "some argue", "both sides". All four counts are also normalised per 100 words, so that longer responses do not automatically look more hedged just because they have more words. From these counts we then assign each response one strategy label, by a priority rule: **refusal** if refusal count is at least 1; otherwise **balanced** if balanced count is at least 2; otherwise **hedged** if hedge count is at least 3; otherwise **direct** if direct count is at least 1; otherwise **mixed**. Refusals come first because they are the strongest signal. Balanced framings come above hedging because explicit two-sided structure is a more committed move than spread-out uncertainty. Because balanced is checked before hedged, responses that satisfy both conditions are merged into the balanced category, which is why the strategy counts and the raw hedge counts can differ. The thresholds were chosen manually after inspecting a small subset of responses and are not learned.

Besides the keyword classifier we use a pre-trained transformer [3] from the BERT family [4]. Specifically, we use the `cardiffnlp/twitter-roberta-base-sentiment-latest` model from the Hugging Face hub, a RoBERTa classifier [5] fine-tuned on about 124 million tweets as part of the TweetEval benchmark [6]. The model returns three probabilities ("positive", "negative", "neutral") that sum to one, and assigns a discrete label set to the class with the highest probability. The explicit neutral class was important: ethical responses are often actually neutral in tone, and a two-class model would force them into an artificial binary split. One note: the model was trained on tweets, where positive and negative mean approval or disapproval. Applied to ethical reasoning, these labels capture emotional register: "positive" responses use reassuring language, "negative" responses use concerned, critical, or cautious language.

A natural alternative to our keyword classifier would be to use an LLM as a judge. We chose not to: using one model to classify another model's outputs adds a second source of model bias, keyword rules are fully transparent, and with a sample size in the tens, a learned classifier cannot be properly trained or validated. The cost is lower sensitivity to paraphrase and some risk of false positives, which we show in Section 3.5.

The pipeline is a Python package, organised by concern. Model access goes through an abstract base class `LLMClient` that defines a single contract: a `query` method that takes a prompt id and prompt text and returns an `LLMResponse`. Concrete clients (such as `GeminiClient`) inherit from it and implement only the provider-specific call. This follows the template method pattern and the open-closed principle: adding a new provider does not require changing existing code. The keyword and sentiment stages are encapsulated in separate `KeywordAnalyzer` and `SentimentAnalyzer` classes, each with its own visualisation module, following the same principle of separation of concerns.

3 Experimental Results

3.1 Dataset Overview

Data collection took three days because the free-tier daily limit is twenty requests per model. The crash-safe restart logic of the experiment runner made this manageable. The final dataset has all sixty successful responses, thirty per model. Average response latency was 15.3 seconds for Flash and 5.7 seconds for Flash-Lite, in line with Flash-Lite's role as the faster variant.

3.2 Keyword-Based Analysis

Figure 1 shows the count of each response strategy by model, and Table 1 reports the mean linguistic features. Of Flash's thirty responses, seventeen (57%) are hedged, twelve (40%) are balanced, and one (3%) is direct. Of Flash-Lite's thirty, sixteen (53%) are balanced, thirteen (43%) are hedged, and one (3%) is a refusal (a false positive shown in Section 3.5). Flash splits about evenly between hedging and balancing, with a single committed answer. Flash-Lite leans more strongly toward explicit two-sided framing.

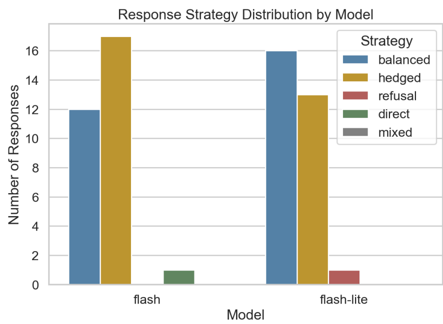

Fig. 1 Distribution of response strategies by model.

Three patterns stand out in the keyword counts. Flash-Lite writes longer responses on average (806 vs. 728 words, +11%) but uses about the same number of sentences: it writes longer sentences, not more of them. Flash-Lite uses about 48% more hedge

Table 1 Mean linguistic features per response, by model.

Feature	Flash	Flash-Lite
Word count	727.5	806.2
Sentence count	45.4	43.5
Hedge markers	10.5	15.6
Refusal markers	0.0	0.03
Direct commitment	1.4	1.0
Balanced framing	1.9	2.4

markers per response (15.6 vs. 10.5). And Flash-Lite is less committal: 31% fewer direct commitment markers and 24% more two-sided markers. So the smaller model is consistently the more linguistically cautious one. Because the two models produce responses of different lengths, we also computed the hedge rate per 100 words: Flash-Lite still hedges more densely (1.98 vs. 1.46 hedges per 100 words), so the pattern is not just an artefact of longer responses.

3.3 Category-Level Analysis

The overall gap is not uniform across categories. Figure 2 shows the mean hedge-marker rate (per 100 words) per category by model. In five of the six categories Flash-Lite hedges more, with the largest gaps on medical ethics, resource allocation, and privacy vs. safety. On speech vs. harm the pattern flips: Flash hedges slightly more. The hedging gap seems to track how personally sensitive a topic is, not how philosophically complex it is.

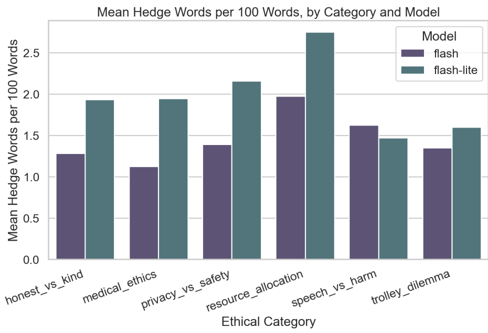

Fig. 2 Mean hedge-marker rate per 100 words, per category and model.

3.4 Sentiment Analysis

We applied the sentiment classifier to all sixty responses. Each response received three probabilities and one discrete label, set to the class with the highest probability.

Averaged across the sixty responses, Flash has a mean negative sentiment of 0.405, against 0.362 for Flash-Lite, a difference of 4.3 percentage points in the direction of Flash being more emotionally concerned. Neutral probabilities go the other way (0.559 for Flash, 0.588 for Flash-Lite). Positive probabilities are low for both models (0.036

and 0.050). Looking at discrete labels, Flash produced nine negative-labelled responses and twenty-one neutral ones, against five negative and twenty-five neutral for Flash-Lite. Neither model produced a single positive-labelled response. Figure 3 shows the mean sentiment probabilities by model.

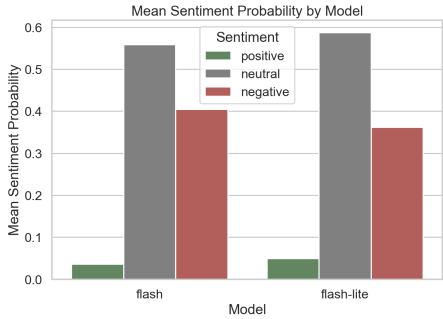

Fig. 3 Mean sentiment probabilities by model.

The sentiment data complicates a simple Flash-Lite is more cautious reading. Flash-Lite does hedge more in language, but Flash adopts a more emotionally concerned tone. So the two classifiers together tell a more detailed story than either alone: the keyword classifier shows Flash-Lite's linguistic caution through hedging and balanced framing, while the sentiment classifier shows Flash's emotional caution through concerned register. The complete absence of positive-labelled responses is itself a finding: both Gemini variants discuss ethical dilemmas without ever using a positive emotional register.

3.5 Example Case Studies

We present two case studies from prompts where the two models differed most sharply.

Case 1: Commit early, or never commit. The prompt `privacy_01` asks whether governments should monitor private messaging to prevent terrorism. Flash takes a clear position early: it argues that governments should not be allowed to monitor all private messaging. This is the single direct-labelled response for Flash. Flash-Lite, on the same prompt, only arrives at a position after six hundred words, under an "if forced to defend a position" heading. Flash treats a question as something to take a position on. Flash-Lite treats it as something to map completely before hesitantly committing.

Case 2: A keyword classifier limitation. The prompt `privacy_02` asks whether to inform a friend that their spouse is cheating. This is Flash-Lite's single refusal label, but it is a false positive: the classifier was triggered by "I can't stand by and watch this dishonesty continue", where "I can't" expresses moral resolve rather than refusal. We keep this case because it illustrates a main limitation of keyword-based classification. The true number of refusals in the corpus is therefore zero.

4 Concluding Remarks

4.1 Summary of Findings

Across sixty valid responses, the keyword classifier shows that the smaller Flash-Lite model uses 48% more hedge markers per response, 24% more two-sided framings, and 31% fewer direct commitment markers, and produces longer responses (806 vs. 728 words). The corpus contains zero true complete refusals. Flash splits roughly evenly between hedging and balancing (with one committed answer), while Flash-Lite mostly balances through explicit two-sided framing. The pattern holds in five of the six categories. The sentiment classifier tells a complementary story: while Flash-Lite hedges more linguistically, Flash uses a more emotionally concerned tone, with mean negative sentiment of 0.405 against 0.362 for Flash-Lite. Neither model produced a single positive-labelled response. So the two models are cautious in different ways, rather than one being uniformly more cautious than the other.

4.2 Interpretation

Two readings are worth stating, neither established by the data alone. First, the differences are not artefacts of capability. You might expect the smaller model to produce shorter responses; the opposite is observed. This fits the idea that the smaller model has been tuned toward a more linguistically cautious register. The sentiment data adds complexity: the larger Flash is the one with the more emotionally concerned tone. So caution is not a single axis but a many-sided construct that the two variants realise differently.

Second, the gap tracks how personally sensitive a topic is, not how philosophically complex it is. Both models behave almost identically on abstract trolley problems and differ most on personal scenarios with real stakes. This is a finding about the register the models adopt, not about how they reason about ethics.

4.3 Limitations

Sixty responses across two models is a small corpus. We report effect sizes rather than statistical significance. LLM outputs are also stochastic; a more careful protocol would query each pair multiple times. Our design chose breadth over depth.

The keyword classifier can both undercount and produce false positives (Section 3.5). We expect these errors to affect both models uniformly and therefore to weaken rather than invent the differences.

The RoBERTa sentiment model has a 512-token input limit. Because the average response length is close to 730–806 words (approximately 970–1070 tokens), the classifier processed roughly the first half of most responses. So the reported sentiment scores reflect the opening register of each response rather than its complete emotional flow. A more detailed approach would chunk long responses and aggregate scores across chunks, which is noted as future work.

The sentiment classifier was fine-tuned on tweets, not on ethical reasoning text. Its labels capture emotional register rather than moral approval, and the sentiment results should be read accordingly.

Comparing two variants of one family controls for provider but means the findings say nothing about cross-provider differences. The prompts also reflect a single researcher's view of what counts as ethical ambiguity.

4.4 Future Work

Future work could scale the dataset, query each model multiple times per prompt, test statistical significance, compare across providers, and evaluate the keyword labels against a small human-coded subset. A more detailed sentiment analysis could also chunk long responses and aggregate sentiment scores across the full text. The pipeline itself is reusable for other research questions with minimal changes.

AI Usage Disclaimer

In line with the course policy, I used OpenAI's ChatGPT (GPT-5.5) during this project. The main use was support with coding, particularly debugging when something was not working, and some editorial help in editing and polishing parts of the written text. The research question, the design of the prompts, the choice of methodology, the analysis, and the interpretation of the findings are my own work. I reviewed and tested every AI-assisted output before keeping it, and I take full responsibility for the final content of the paper and the code.

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